

App: app Grade

file : main.dart

```
lib > main.dart > ...
1  import 'package:flutter/material.dart';
2  import 'home.dart';
3
Run | Debug | Profile
4  void main() {
5    runApp(const MyApp());
6  }
7
8  class MyApp extends StatelessWidget {
9    const MyApp({super.key});
10
11    @override
12    Widget build(BuildContext context) {
13      return MaterialApp(
14        title: 'Flutter Demo',
15        theme: ThemeData(colorScheme: .fromSeed(seedColor: Colors.deepPurple)),
16        debugShowCheckedModeBanner: false,
17        home: MyHomePage(),
18      ); // MaterialApp
19    }
20  }
21
```

5:29

App Grade

Firstname
Phasinee

Lastname
Kengklang

Field of Study
☐ INE ☒ INET

Subject
COMP NET DESIGN & IMPLEMENT

Midterm Exam
20

Final Exam
22

Classwork score
42

Calculate Grade

5:29

← App Grade

Pasinee Kengklang
Information and Network Engineering (INET)

A
COMP NET DESIGN & IMPLEMENT

Score Total : 84.0

Midterm : 20
Final : 22
Classwork : 42

Back to Form

App: app Grade

file : home.dart

```
lib > home.dart > _MyHomePageState > build
1  import 'package:flutter/material.dart';
2  import 'dart:ui';
3  import 'result.dart';
4
5  enum Department { INE, INET }
6
7  class MyHomePage extends StatefulWidget {
8    const MyHomePage({super.key});
9
10   @override
11   State<MyHomePage> createState() => _MyHomePageState();
12 }
13
14 class _MyHomePageState extends State<MyHomePage> {
15   TextEditingController firstname = TextEditingController();
16   TextEditingController lastname = TextEditingController();
17   TextEditingController mid_score = TextEditingController();
18   TextEditingController final_score = TextEditingController();
19   TextEditingController class_score = TextEditingController();
20
21   String? _selectedItem;
22   List<String> _dropdownmenu = [
23     'MOBILE APPLICATION DEVELOPMENT',
24     'FULL STACK SYSTEM DEVELOPMENT',
25     'COMP NET DESIGN & IMPLEMENT',
26     'DATABASE SYSTEM',
27   ];
28
29   final num1con = TextEditingController();
30   final _formKey = GlobalKey<FormState>();
31   Department? _department;
32
33   void initState() {
34     super.initState();
35     _selectedItem = _dropdownmenu[0];
36   }
37
38   @override
39   Widget build(BuildContext context) {
40     return Scaffold(
41       appBar: AppBar(
42         title: Text('App Grade'),
43         backgroundColor: Colors.green,
44         foregroundColor: Colors.white,
45       ), // AppBar
46       body: Padding(
47         padding: EdgeInsets.all(20),
```

```

46 body: Padding(
47   padding: EdgeInsets.all(20),
48   child: Form(
49     key: _formKey,
50     child: SingleChildScrollView(
51       child: Column(
52         crossAxisAlignment: CrossAxisAlignment.start,
53         children: [
54           TextInputName(
55             label: 'Firstname',
56             control: firstname,
57             icon: Icons.person,
58             validator: (val) {
59               if (val == null || val.isEmpty) {
60                 return 'Please Enter Firstname';
61               }
62               return null;
63             },
64           ),
65           SizedBox(height: 10),
66           TextInputName(
67             label: 'Lastname',
68             control: lastname,
69             icon: Icons.person,
70             validator: (val) {
71               if (val == null || val.isEmpty) {
72                 return 'Please Enter Lastname';
73               }
74               return null;
75             },
76           ),
77           SizedBox(height: 20),
78           Text(
79             "Field of Study",
80             style: TextStyle(
81               fontSize: 16,
82               fontWeight: FontWeight.bold,
83               color: Colors.grey[800],
84             ), // TextStyle
85           ), // Text
86           SizedBox(height: 5),
87           FormField<Department>(
88             validator: (value) {
89               if (_department == null) {
90                 return 'Please choose Department';
91               }
92               return null;
93             },
94           builder: (state) {
95             return Column(
96               crossAxisAlignment: CrossAxisAlignment.start,
97               children: [
98                 Row(
99                   children: [

```

```

100 Expanded(
101     child: Container(
102         margin: const EdgeInsets.all(4.0),
103         decoration: BoxDecoration(
104             border: Border.all(
105                 color: Colors.green.shade200,
106                 width: 1.5,
107             ), // Border.all
108             borderRadius: BorderRadius.circular(8.0),
109         ), // BoxDecoration
110         child: RadioListTile<Department>(
111             title: Text(
112                 Department.INE.name,
113                 style: TextStyle(fontSize: 18),
114             ), // Text
115             value: Department.INE,
116             groupValue: _department,
117             dense: true,
118             contentPadding: EdgeInsets.zero,
119             onChanged: (val) {
120                 setState(() {
121                     _department = val;
122                 });
123             },
124         ), // RadioListTile
125     ), // Container
126 ), // Expanded
127 Expanded(
128     child: Container(
129         margin: const EdgeInsets.all(4.0),
130         decoration: BoxDecoration(
131             border: Border.all(
132                 color: Colors.green.shade200,
133                 width: 1.5,
134             ), // Border.all
135             borderRadius: BorderRadius.circular(8.0),
136         ), // BoxDecoration
137         child: RadioListTile<Department>(
138             title: Text(
139                 Department.INET.name,
140                 style: TextStyle(fontSize: 18),
141             ), // Text
142             value: Department.INET,
143             groupValue: _department,
144             dense: true,
145             contentPadding: EdgeInsets.zero,
146             onChanged: (val) {
147                 setState(() {
148                     _department = val;
149                 });
150             },
151         ), // RadioListTile
152     ), // Container
153 ), // Expanded
154 ],
155 ), // Row

```



```

156         if (state.hasError)
157         |
158         |   Padding(
159         |   |   padding: const EdgeInsets.only(left: 12, top: 5),
160         |   |   child: Text(
161         |   |   |   state.errorText!,
162         |   |   |   style: TextStyle(
163         |   |   |   |   color: Colors.red[700],
164         |   |   |   |   fontSize: 12,
165         |   |   |   ), // TextStyle
166         |   |   ), // Text
167         |   ), // Padding
168         |   ],
169         |   ); // Column
170         }, // FormField
171         SizedBox(height: 10),
172         Text(
173         |   "Subject",
174         |   style: TextStyle(
175         |   |   fontSize: 16,
176         |   |   fontWeight: FontWeight.bold,
177         |   |   color: Colors.grey[800],
178         |   ), // TextStyle
179         ), // Text
180         SizedBox(height: 5),
181         Container(
182         |   padding: EdgeInsets.symmetric(horizontal: 10, vertical: 5),
183         |   decoration: BoxDecoration(
184         |   |   borderRadius: BorderRadius.circular(10),
185         |   |   border: Border.all(color: Colors.green.shade100, width: 2),
186         |   ), // BoxDecoration
187         |   child: DropdownButtonHideUnderline(
188         |   |   child: DropdownButtonFormField(
189         |   |   |   value: _selectedItem,
190         |   |   |   validator: (val) {
191         |   |   |   |   if (val == null || val.isEmpty) {
192         |   |   |   |   |   return 'Please Choose subject';
193         |   |   |   |   }
194         |   |   |   |   return null;
195         |   |   |   },
196         |   |   |   decoration: InputDecoration(
197         |   |   |   |   border: InputBorder.none,
198         |   |   |   |   enabledBorder: InputBorder.none,
199         |   |   |   |   focusedBorder: InputBorder.none,
200         |   |   |   ), // InputDecoration
201         |   |   |   items: _dropdownmenu
202         |   |   |   |   .map(
203         |   |   |   |   |   (e) => DropdownMenuItem(value: e, child: Text(e)),
204         |   |   |   |   )
205         |   |   |   |   .toList(),
206         |   |   |   onChanged: (val) {
207         |   |   |   |   setState(() {
208         |   |   |   |   |   _selectedItem = val;
209         |   |   |   |   });
210         |   |   |   },
211         |   |   |   isExpanded: true,
212         |   |   ), // DropdownButtonFormField
213         |   ), // DropdownButtonHideUnderline
214         ), // Container
215         SizedBox(height: 20),

```

```

216 TextInputScore(
217   label: 'Midterm Exam',
218   control: mid_score,
219   icon: Icons.quiz_sharp,
220   validator: (val) {
221     if (val == null || val.isEmpty) {
222       return 'Please Enter midterm score';
223     } else if (double.tryParse(val) == null) {
224       return 'Please Enter type number';
225     }
226     return null;
227   },
228 ),
229 SizedBox(height: 10),
230 TextInputScore(
231   label: 'Final Exam',
232   control: final_score,
233   icon: Icons.quiz_sharp,
234   validator: (val) {
235     if (val == null || val.isEmpty) {
236       return 'Please Enter final score';
237     } else if (double.tryParse(val) == null) {
238       return 'Please Enter type number';
239     }
240     return null;
241   },
242 ),
243 SizedBox(height: 10),
244 TextInputScore(
245   label: 'Classwork score',
246   control: class_score,
247   icon: Icons.inventory_2_sharp,
248   validator: (val) {
249     if (val == null || val.isEmpty) {
250       return 'Please Enter classwork score';
251     } else if (double.tryParse(val) == null) {
252       return 'Please Enter type number';
253     }
254     return null;
255   },
256 ),
257 SizedBox(height: 30),
258 ElevatedButton(
259   onPressed: () {
260     if (_formKey.currentState!.validate()) {
261       double m = double.tryParse(mid_score.text) ?? 0;
262       double f = double.tryParse(final_score.text) ?? 0;
263       double c = double.tryParse(class_score.text) ?? 0;
264       double total = m + f + c;
265       if (total > 100) {
266         ScaffoldMessenger.of(context).showSnackBar(
267           const SnackBar(
268             content: Text(
269               'คะแนนรวมไม่ถูกต้อง (เกิน 100 คะแนน)',
270             ), // Text
271             backgroundColor: Colors.red,
272             behavior: SnackBarBehavior.floating,
273             duration: Duration(seconds: 2),
274           ), // SnackBar
275         );

```

```

276     } else {
277         Navigator.push(
278             context,
279             MaterialPageRoute(
280                 builder: (context) {
281                     return MyResult(
282                         fname: firstname.text,
283                         lname: lastname.text,
284                         dept: _department?.name ?? "",
285                         subject: _selectedItem ?? "",
286                         midterm: mid_score.text,
287                         finalScore: final_score.text,
288                         classwork: class_score.text,
289                     ); // MyResult
290                 },
291             ), // MaterialPageRoute
292         );
293     }
294     } else {
295         print("กรุณากรอกข้อมูลให้ครบ");
296     }
297 },
298 style: ElevatedButton.styleFrom(
299     padding: EdgeInsets.all(0),
300     minimumSize: Size(double.infinity, 50),
301     textStyle: TextStyle(
302         fontSize: 20,
303         fontWeight: FontWeight.bold,
304     ), // TextStyle
305     backgroundColor: Colors.green,
306     foregroundColor: Colors.white,
307     elevation: 2,
308     shadowColor: Colors.blue,
309     side: BorderSide(color: Colors.green[800]!, width: 1),
310     shape: StadiumBorder(),
311 ),
312 child: Text('Calculate Grade'),
313 ), // ElevatedButton
314 ],
315 ), // Column
316 ), // SingleChildScrollView
317 ), // Form
318 ), // Padding
319 ); // Scaffold
320 }
321 }

```

```

323 Widget TextInputName({
324   required String label,
325   required TextEditingController control,
326   required IconData icon,
327   String? Function(String?)? validator,
328 }) {
329   return TextFormField(
330     controller: control,
331     validator: validator,
332     keyboardType: TextInputType.text,
333     style: TextStyle(fontSize: 20, fontWeight: FontWeight.w500),
334     decoration: InputDecoration(
335       labelText: label,
336       prefixIcon: Icon(icon, size: 30),
337       focusedBorder: OutlineInputBorder(
338         borderRadius: BorderRadius.circular(10),
339         borderSide: const BorderSide(color: Colors.green, width: 2.0),
340       ), // OutlineInputBorder
341       enabledBorder: OutlineInputBorder(
342         borderRadius: BorderRadius.circular(10),
343         borderSide: BorderSide(color: Colors.green.shade100, width: 2.0),
344       ), // OutlineInputBorder
345       errorBorder: OutlineInputBorder(
346         borderRadius: BorderRadius.circular(10),
347         borderSide: const BorderSide(color: Colors.red, width: 2.0),
348       ), // OutlineInputBorder
349       focusedErrorBorder: OutlineInputBorder(
350         borderRadius: BorderRadius.circular(10),
351         borderSide: const BorderSide(color: Colors.red, width: 2.5),
352       ), // OutlineInputBorder
353     ), // InputDecoration
354   ); // TextFormField
355 }

```

```

357 Widget TextInputScore({
358   required String label,
359   required TextEditingController control,
360   required IconData icon,
361   String? Function(String?)? validator,
362 }) {
363   return TextFormField(
364     controller: control,
365     validator: validator,
366     keyboardType: TextInputType.number,
367     style: TextStyle(fontSize: 20, fontWeight: FontWeight.w500),
368     decoration: InputDecoration(
369       labelText: label,
370       prefixIcon: Icon(icon),
371       focusedBorder: OutlineInputBorder(
372         borderRadius: BorderRadius.circular(10),
373         borderSide: const BorderSide(color: Colors.green, width: 2.0),
374       ), // OutlineInputBorder
375       enabledBorder: OutlineInputBorder(
376         borderRadius: BorderRadius.circular(10),
377         borderSide: BorderSide(color: Colors.green.shade100, width: 2.0),
378       ), // OutlineInputBorder
379       errorBorder: OutlineInputBorder(
380         borderRadius: BorderRadius.circular(10),
381         borderSide: const BorderSide(color: Colors.red, width: 2.0),
382       ), // OutlineInputBorder
383       focusedErrorBorder: OutlineInputBorder(
384         borderRadius: BorderRadius.circular(10),
385         borderSide: const BorderSide(color: Colors.red, width: 2.5),
386       ), // OutlineInputBorder
387     ), // InputDecoration
388   ); // TextFormField
389 }

```



```
lib > result.dart > MyResult > build
1  import 'package:flutter/material.dart';
2  import 'dart:ui';
3
4  class MyResult extends StatelessWidget {
5    const MyResult({
6      super.key,
7      required this.fname,
8      required this.lname,
9      required this.dept,
10     required this.subject,
11     required this.midterm,
12     required this.finalScore,
13     required this.classwork,
14   });
15
16   final String fname;
17   final String lname;
18   final String dept;
19   final String subject;
20   final String midterm;
21   final String finalScore;
22   final String classwork;
23   @override
24   Widget build(BuildContext context) {
25     double m = double.tryParse(midterm) ?? 0;
26     double f = double.tryParse(finalScore) ?? 0;
27     double c = double.tryParse(classwork) ?? 0;
28     final double total = m + f + c;
29     String grade = "";
30
31     if (total >= 80) {
32       grade = 'A';
33     } else if (total >= 75) {
34       grade = 'B+';
35     } else if (total >= 70) {
36       grade = 'B';
37     } else if (total >= 65) {
38       grade = 'C+';
39     } else if (total >= 60) {
40       grade = 'C';
41     } else if (total >= 55) {
42       grade = 'D+';
43     } else if (total >= 50) {
44       grade = 'D';
45     } else {
46       grade = 'F';
47     }
```

```

48 return Scaffold(
49   appBar: AppBar(
50     title: Text('App Grade'),
51     backgroundColor: Colors.green,
52     foregroundColor: Colors.white,
53   ), // AppBar
54   body: Padding(
55     padding: EdgeInsets.all(10),
56     child: Column(
57       children: [
58         Container(
59           padding: EdgeInsets.symmetric(vertical: 10, horizontal: 10),
60           width: double.infinity,
61           decoration: BoxDecoration(
62             color: Colors.white,
63             border: Border.all(color: Colors.green, width: 1),
64             borderRadius: BorderRadius.circular(5),
65             boxShadow: [
66               BoxShadow(
67                 color: Colors.grey[200]!,
68                 spreadRadius: 2,
69                 blurRadius: 10,
70                 offset: const Offset(0, 5),
71               ), // BoxShadow
72             ],
73           ), // BoxDecoration
74           child: Column(
75             crossAxisAlignment: CrossAxisAlignment.center,
76             children: [
77               Text(
78                 'Pasinee Kengklang',
79                 style: const TextStyle(
80                   fontSize: 26,
81                   fontWeight: FontWeight.w500,
82                 ), // TextStyle
83               ), // Text
84               Text('Information and Network Engineering ($dept)'),
85             ],
86           ), // Column
87         ), // Container
88         SizedBox(height: 25),
89         Container(
90           padding: EdgeInsets.symmetric(vertical: 10, horizontal: 10),
91           width: double.infinity,
92           decoration: BoxDecoration(
93             color: Colors.white,
94             border: Border.all(color: Colors.green, width: 1),
95             borderRadius: BorderRadius.circular(5),
96             boxShadow: [
97               BoxShadow(
98                 color: Colors.grey[200]!,
99                 spreadRadius: 2,
100                blurRadius: 10,
101                offset: const Offset(0, 5),
102              ), // BoxShadow
103            ],
104          ), // BoxDecoration

```

```

105 child: Column(
106   crossAxisAlignment: CrossAxisAlignment.center,
107   children: [
108     if (total < 50) ...[
109       Container(
110         width: 50,
111         height: 50,
112         alignment: Alignment.center,
113         decoration: BoxDecoration(
114           color: Colors.red[700],
115           border: Border.all(color: Colors.red[700]!),
116           borderRadius: BorderRadius.circular(50),
117         ), // BoxDecoration
118         child: Text(
119           'F',
120           style: const TextStyle(
121             color: Colors.white,
122             fontSize: 24,
123             fontWeight: FontWeight.w500,
124           ), // TextStyle
125         ), // Text
126       ), // Container
127     ] else ...[
128       Container(
129         width: 50,
130         height: 50,
131         alignment: Alignment.center,
132         decoration: BoxDecoration(
133           color: Colors.green[700],
134           border: Border.all(color: Colors.green[700]!),
135           borderRadius: BorderRadius.circular(50),
136         ), // BoxDecoration
137         child: Text(
138           grade.toString(),
139           style: const TextStyle(
140             color: Colors.white,
141             fontSize: 24,
142             fontWeight: FontWeight.w500,
143           ), // TextStyle
144         ), // Text
145       ), // Container
146     ],
147   ), // Column
148   SizedBox(height: 10),
149   Text(
150     '$subject',
151     style: const TextStyle(
152       fontSize: 18,
153       fontWeight: FontWeight.w500,
154     ), // TextStyle
155   ), // Text
156 ], // Column
157 ), // Container
158 ), // Column
159 ), // Container

```

```

160 Container(
161   width: double.infinity,
162   padding: EdgeInsets.symmetric(horizontal: 10, vertical: 10),
163   decoration: BoxDecoration(
164     color: Colors.white,
165     border: Border.all(color: Colors.green),
166     borderRadius: BorderRadius.circular(5),
167   ), // BoxDecoration
168   child: Column(
169     crossAxisAlignment: CrossAxisAlignment.start,
170     children: [
171       Row(
172         children: [
173           Icon(Icons.bar_chart_sharp, size: 20),
174           SizedBox(width: 2),
175           Text(
176             'Score Total : $total',
177             style: TextStyle(
178               color: Colors.black,
179               fontSize: 18,
180               fontWeight: FontWeight.w500,
181             ), // TextStyle
182           ), // Text
183         ],
184       ), // Row
185       Divider(color: Colors.green[200]),
186       SizedBox(height: 5),
187       Row(
188         children: [
189           Icon(Icons.quiz_sharp, size: 20),
190           SizedBox(width: 2),
191           Text(
192             "Midterm : $midterm",
193             style: TextStyle(
194               fontSize: 16,
195               fontWeight: FontWeight.w400,
196             ), // TextStyle
197           ), // Text
198         ],
199       ), // Row
200       SizedBox(height: 5),
201       Row(
202         children: [
203           Icon(Icons.quiz_sharp, size: 20),
204           SizedBox(width: 2),
205           Text(
206             "Final : $finalScore",
207             style: TextStyle(
208               fontSize: 16,
209               fontWeight: FontWeight.w400,
210             ), // TextStyle
211           ), // Text
212         ],
213       ), // Row
214       SizedBox(height: 5),

```

```

216         Row(
217           children: [
218             Icon(Icons.inventory_2_sharp, size: 20),
219             SizedBox(width: 2),
220             Text(
221               "Classwork : $classwork",
222               style: TextStyle(
223                 fontSize: 16,
224                 fontWeight: FontWeight.w400,
225               ), // TextStyle
226             ), // Text
227           ],
228         ), // Row
229       ],
230     ), // Column
231   ), // Container
232   SizedBox(height: 40),
233   ElevatedButton(
234     onPressed: () {
235       Navigator.pop(context);
236     },
237     style: ElevatedButton.styleFrom(
238       minimumSize: Size(double.infinity, 50),
239       padding: EdgeInsets.symmetric(horizontal: 10),
240       foregroundColor: Colors.white,
241       backgroundColor: Colors.green,
242       elevation: 5,
243     ),
244     child: Text('Back to Form', style: TextStyle(fontSize: 20)),
245   ), // ElevatedButton
246 ],
247 ), // Column
248 ), // Padding
249 ); // Scaffold
250 }
251 }
252

```